MANIPAL INSTITUTE OF TECHNOLOGY
BENGALURU
(A constituent unit of MAHE, Manipal)

VII Semester ENDTERM TEST
Machine Learning (CSE_4032)

Time Duration: 3 Hours **Date: 20-Nov-2025** **Max marks: 50 MARKS**

Q. No.	Topic	Marks	BL	CO
1 A.	Apply the traditional K-Means clustering (partitional) algorithm with Euclidean distance to the following 2D dataset: (see table below) With k=2, use P2 and P6 as the initial cluster centers.	5	Apply	CO. 4

Point	Coordinates (x, y)
P1	(1, 2)
P2	(1, 4)
P3	(2, 3)
P4	(6, 8)
P5	(7, 9)
P6	(8, 7)

Ans.

k = 2, initial centers:

- Cluster 1 center $C1^{(0)} = P2 = (1, 4)$
- Cluster 2 center $C2^{(0)} = P6 = (8, 7)$

Distance used: Euclidean

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Compute distances to initial centers

To C1 = (1,4)

- P1: $d = \sqrt{(1-1)^2 + (2-4)^2} = \sqrt{4} = 2.0$
- P2: $d = \sqrt{(1-1)^2 + (4-4)^2} = 0$
- P3: $d = \sqrt{(2-1)^2 + (3-4)^2} = \sqrt{1+1} = \sqrt{2} \approx 1.41$
- P4: $d = \sqrt{(6-1)^2 + (8-4)^2} = \sqrt{25+16} = \sqrt{41} \approx 6.40$
- P5: $d = \sqrt{(7-1)^2 + (9-4)^2} = \sqrt{36+25} = \sqrt{61} \approx 7.81$
- P6: $d = \sqrt{(8-1)^2 + (7-4)^2} = \sqrt{49+9} = \sqrt{58} \approx 7.62$

To C2 = (8,7)

- P1: $d = \sqrt{(1-8)^2 + (2-7)^2} = \sqrt{49+25} = \sqrt{74} \approx 8.60$
- P2: $d = \sqrt{(1-8)^2 + (4-7)^2} = \sqrt{49+9} = \sqrt{58} \approx 7.62$
- P3: $d = \sqrt{(2-8)^2 + (3-7)^2} = \sqrt{36+16} = \sqrt{52} \approx 7.21$
- P4: $d = \sqrt{(6-8)^2 + (8-7)^2} = \sqrt{4+1} = \sqrt{5} \approx 2.24$
- P5: $d = \sqrt{(7-8)^2 + (9-7)^2} = \sqrt{1+4} = \sqrt{5} \approx 2.24$
- P6: $d = \sqrt{(8-8)^2 + (7-7)^2} = 0$

2. Assign each point to nearest center

Compare distances:

- P1: 2.0 (C1) vs 8.60 (C2) → **Cluster 1**

- P2: 0 (C1) vs 7.62 (C2) → **Cluster 1**
- P3: 1.41 (C1) vs 7.21 (C2) → **Cluster 1**
- P4: 6.40 (C1) vs 2.24 (C2) → **Cluster 2**
- P5: 7.81 (C1) vs 2.24 (C2) → **Cluster 2**
- P6: 7.62 (C1) vs 0 (C2) → **Cluster 2**

So after Iteration 1:

- **Cluster 1**: {P1, P2, P3}
- **Cluster 2**: {P4, P5, P6}

Recompute centroids

Cluster 1 centroid $C1^{(1)}$ (P1, P2, P3):

$$x_{C1} = \frac{1+1+2}{3} = \frac{4}{3} \approx 1.33, y_{C1} = \frac{2+4+3}{3} = \frac{9}{3} = 3$$

So

$$C1^{(1)} = \left(\frac{4}{3}, 3\right) \approx (1.33, 3.00)$$

Cluster 2 centroid $C2^{(1)}$ (P4, P5, P6):

$$x_{C2} = \frac{6+7+8}{3} = \frac{21}{3} = 7, y_{C2} = \frac{8+9+7}{3} = \frac{24}{3} = 8$$

So

$$C2^{(1)} = (7,8)$$

Iteration 2

Use new centers $C1^{(1)}$ = (4/3, 3), $C2^{(1)}$ = (7, 8).

(Values below are approximated.)

Distances to $C1^{(1)}$ ≈ (1.33, 3):

- P1 (1,2): $d \approx 1.05$
- P2 (1,4): $d \approx 1.05$
- P3 (2,3): $d \approx 0.67$
- P4 (6,8): $d \approx 6.84$
- P5 (7,9): $d \approx 8.25$
- P6 (8,7): $d \approx 7.77$

Distances to $C2^{(1)}$ = (7,8):

- P1: $d \approx 8.49$
- P2: $d \approx 7.21$
- P3: $d \approx 7.07$
- P4: $d = 1.0$
- P5: $d = 1.0$
- P6: $d \approx 1.41$

Assignments again:

- P1, P2, P3 closer to C1 → **Cluster 1**
- P4, P5, P6 closer to C2 → **Cluster 2**

Cluster memberships **did not change**.

Recomputing the centroids will give the same centers (4/3, 3) and (7, 8).

So the algorithm has **converged**.

Final Answer

- **Final clusters:**
  - **Cluster 1**: {P1(1,2), P2(1,4), P3(2,3)}
  - **Cluster 2**: {P4(6,8), P5(7,9), P6(8,7)}
- **Final cluster centers (centroids):**
  - **Center of Cluster 1**:

$$\left(\frac{4}{3}, 3\right) \approx (1.33,\ 3.00)$$

- **Center of Cluster 2**:

$$(7,\ 8)$$

Topic covered	Marks
Correctly stating initial centers	0.5
Correct assignment of 6 points (P1 to P6)	0.5*6=3
Correct re-computation of centroids after Iteration 1	0.5*2= 1
Clearly stating final clusters and centroids	0.5

1 B.	Spectral clustering involves constructing a similarity graph and computing the graph Laplacian. Analyze the role of each of the following in determining cluster boundaries: (a) the degree matrix D, (b) the adjacency (similarity) matrix W, and (c) the Laplacian $L = D - W$.	5	Analysis	CO. 4

Ans.

Topic covered	**Marks**
Degree Matrix D	
• Mentions that D_{ii} measures total similarity / local density	0.5
• Explains how degree affects normalization or influence of nodes	0.5
• Explains role in revealing boundary nodes with low degree	0.5
Adjacency Matrix W	
• States that W encodes similarities / connectivity	0.5
• Notes that strong edges retain within-cluster ties	0.5
• Notes that weak edges reduce cross-cluster influence	0.5
Laplacian $L = D - W$–	
• Explains Laplacian's role in graph-cut / boundary detection	0.5
• States smoothness effect	0.5
• Explains eigenvectors creating separable embedding	0.5
• Clearly links Laplacian behavior to final cluster boundaries	0.5

2 A.	You are given data on alien specimens observed in a laboratory. The goal is to predict whether an alien is Hostile (Yes) or Peaceful (No) based on its ToxinLevel reading.	5	Apply	CO. 5

ID	ToxinLevel	Hostile
1	1	No
2	2	No
3	3	No
4	7	Yes
5	8	Yes
6	9	Yes

Apply the CART algorithm using the Gini index to construct a univariate decision tree that predicts whether an alien is *Hostile*.

Gini at the Root Node

At root:

- Hostile = Yes: 3 (IDs 4,5,6)
- Hostile = No: 3 (IDs 1,2,3)

$$p(\text{Yes}) = \frac{3}{6} = 0.5, p(\text{No}) = 0.5$$
$$\text{Gini(root)} = 1 - (0.5^2 + 0.5^2) = 1 - (0.25 + 0.25) = 0.5$$

Candidate split points (midpoints between consecutive values):

- Between 1 and 2 → **1.5**
- Between 2 and 3 → **2.5**
- Between 3 and 7 → **5**
- Between 7 and 8 → **7.5**
- Between 8 and 9 → **8.5**

Split at ToxinLevel ≤ 1.5

Left node (≤ 1.5)

- Class counts: No = 1, Yes = 0

$$p(\text{No}) = 1, p(\text{Yes}) = 0$$
$$\text{Gini}_L = 1 - (1^2 + 0^2) = 1 - 1 = 0$$

Right node (> 1.5)

- Class counts: No = 2, Yes = 3, total = 5

$$p(\text{No}) = \frac{2}{5} = 0.4, p(\text{Yes}) = \frac{3}{5} = 0.6$$
$$\text{Gini}_R = 1 - (0.4^2 + 0.6^2) = 1 - (0.16 + 0.36) = 1 - 0.52 = 0.48$$

Weighted Gini

$$N_L = 1, N_R = 5, N = 6$$
$$\text{Gini(split at 1.5)} = \frac{1}{6} \cdot 0 + \frac{5}{6} \cdot 0.48 = 0 + 0.4 = 0.40$$

Split at ToxinLevel ≤ 2.5

Left node (≤ 2.5)

- Class counts: No = 2, Yes = 0

$$p(\text{No}) = 1, p(\text{Yes}) = 0$$
$$\text{Gini}_L = 1 - (1^2 + 0^2) = 0$$

Right node (> 2.5)

- Class counts: No = 1, Yes = 3, total = 4

$$p(\text{No}) = \frac{1}{4} = 0.25, p(\text{Yes}) = \frac{3}{4} = 0.75$$
$$\text{Gini}_R = 1 - (0.25^2 + 0.75^2) = 1 - (0.0625 + 0.5625) = 1 - 0.625 = 0.375$$

Weighted Gini

$$N_L = 2, N_R = 4$$
$$\text{Gini(split at 2.5)} = \frac{2}{6} \cdot 0 + \frac{4}{6} \cdot 0.375 = 0 + \frac{2}{3} \cdot 0.375 = 0.25$$

Split at ToxinLevel ≤ 5

Left node (≤ 5)

- Class counts: No = 3, Yes = 0

$$p(\text{No}) = 1, p(\text{Yes}) = 0$$
$$\text{Gini}_L = 1 - (1^2 + 0^2) = 0$$

Right node (> 5)

- Class counts: Yes = 3, No = 0

$$p(\text{Yes}) = 1, p(\text{No}) = 0$$
$$\text{Gini}_R = 1 - (1^2 + 0^2) = 0$$

Weighted Gini

$$N_L = 3, N_R = 3$$
$$\text{Gini(split at 5)} = \frac{3}{6} \cdot 0 + \frac{3}{6} \cdot 0 = 0$$

Split at ToxinLevel ≤ 7.5

Left node (≤ 7.5)

- Class counts: No = 3, Yes = 1, total = 4

$$p(\text{No}) = \frac{3}{4} = 0.75, p(\text{Yes}) = \frac{1}{4} = 0.25$$
$$\text{Gini}_L = 1 - (0.75^2 + 0.25^2) = 1 - (0.5625 + 0.0625) = 1 - 0.625 = 0.375$$

Right node (> 7.5)

- Class counts: Yes = 2, No = 0

$$p(\text{Yes}) = 1, p(\text{No}) = 0$$
$$\text{Gini}_R = 1 - (1^2 + 0^2) = 0$$

Weighted Gini

$$N_L = 4, N_R = 2$$
$$\text{Gini(split at 7.5)} = \frac{4}{6} \cdot 0.375 + \frac{2}{6} \cdot 0 = \frac{2}{3} \cdot 0.375 = 0.25$$

Split at ToxinLevel ≤ 8.5

Left node (≤ 8.5)

- Class counts: No = 3, Yes = 2, total = 5

$$p(\text{No}) = \frac{3}{5} = 0.6, p(\text{Yes}) = \frac{2}{5} = 0.4$$
$$\text{Gini}_L = 1 - (0.6^2 + 0.4^2) = 1 - (0.36 + 0.16) = 1 - 0.52 = 0.48$$

Right node (> 8.5)

- Class counts: Yes = 1, No = 0

$$\text{Gini}_R = 1 - (1^2 + 0^2) = 0$$

Weighted Gini

$$N_L = 5, N_R = 1$$
$$\text{Gini(split at 8.5)} = \frac{5}{6} \cdot 0.48 + \frac{1}{6} \cdot 0 = 0.4$$

Best split = ToxinLevel ≤ 5 (minimum Gini = 0).

So final CART rule:

- If **ToxinLevel ≤ 5 → Hostile = No**
- If **ToxinLevel > 5 → Hostile = Yes**

Topic covered	Marks
Root Node Gini	0.5
Identification of Candidate Splits	0.5
Weighted Gini Calculations for Candidate Splits	0.5*2=2.5
Correct Identification of Best Split	0.5
Final Tree Construction ToxinLevel ≤ 5 → Hostile = No ToxinLevel > 5 → Hostile = Yes	 0.5 0.5

2 B.	Examine the relationship between the depth of a fully grown decision tree, its fit-on training data, and its performance on unseen data, and analyze the impact of this relationship on the model's generalization ability.	3	Analysis	CO. 5

Ans.	

Topic covered	Marks
Depth vs Training Fit • States that deeper trees fit training data extremely well / nearly zero training error • Mentions memorization / overfitting due to high depth	0.5 0.5
Depth vs Unseen/Test Data Performance • States that fully grown deep trees perform poorly on new data / high variance • Mentions reason: overly specific boundaries / noise fitting	0.5
. Impact on Generalization Ability • Correctly describes bias–variance trade-off across tree depths (low depth = underfit, high depth = overfit) • States that controlled depth or pruning improves generalization	0.5 0.5

2 C.	Two attributes, A and B, both produce the same Information Gain when splitting a dataset. However, attribute A has 2 possible values, and B has 10 possible values. Explain which attribute a decision tree should prefer and why, even though both have the same Information Gain.	2	Understand	CO. 5
Ans.				

Topic covered	Marks
States that the decision tree should prefer **A (with 2 values)**.	0.5
Explains that attributes with many values (like B with 10 values) tend to **overfit** or produce overly specific splits.	0.5
Notes that fewer branches → **simpler tree**, better generalization.	0.5
Mentions that Information Gain is biased toward attributes with many values and therefore needs correction, so A is preferred despite equal IG.	0.5

3 A.	Examine the influence of membership weights in Fuzzy K-Means on the transformation of hard cluster assignments in standard K-Means and analyze their impact on interpreting boundary points between clusters.	4	Analysis	CO. 4
Ans.				

Topic covered	Marks
Hard vs. fuzzy assignment distinction : States that K-Means uses hard assignments while Fuzzy K-Means uses membership weights.	0.5
Explanation of weighted contribution to centroids : Mentions that centroids are updated using weighted averages based on membership values.	0.5
Transformation effect on clustering behavior: Explains that fuzzy assignments soften rigid cluster boundaries of standard K-Means.	0.5
Interpretation of boundary points: uncertainty : States that boundary points get comparable membership values for multiple clusters.	0.5
Identification of overlapping regions: Notes that fuzzy memberships highlight transitional or overlapping cluster areas.	0.5
Reduced misclassification / better boundary handling Mentions that partial memberships prevent forcing ambiguous points into a single cluster.	0.5
Improved interpretability: Explains that membership weights give richer information about cluster alignment.	0.5
Coherent concluding insight about overall impact	

	Summarizes that membership weights produce smoother boundaries and clearer interpretation.			
3 B.	Explain the concepts of the Bias–Variance Tradeoff, Occam's Razor, and the No Free Lunch Theorem, and describe the way these principles collectively represent the challenges involved in selecting a model that generalizes effectively. Include one supporting example.	3	Understand	CO. 1

Topic covered	Marks
Bias–Variance Tradeoff Explanation • States that underfitting comes from high bias and overfitting from high variance, and that a balance is needed for good generalization.	0.5
Occam's Razor Explanation • States that, among models with similar performance, the simpler model is preferred because it generalizes better.	0.5
No Free Lunch Theorem Explanation • States that no single model is best for all datasets; model performance depends on the specific problem	0.5
Combined Interpretation / Collective Challenge • Explains how these principles together show that model selection is a tradeoff involving simplicity, complexity, and dataset-dependence.	0.5
Example Provided • **Gives any correct example illustrating overfitting, underfitting, or model selection influenced by these principles.**	0.5
Clear linkage to generalization • **States explicitly that these principles relate to choosing a model that generalizes well beyond the training data**	0.5

3 C.	Analyze how boosting's iterative weight-updating strategy strengthens the ensemble compared to using a single weak learner.	3	Analysis	CO. 6

Ans.

Boosting strengthens an ensemble by repeatedly adjusting the weights of training instances so that **misclassified samples receive higher attention** in later rounds. Unlike a single weak learner that focuses on the entire dataset uniformly, boosting forces each successive learner to concentrate on the mistakes of the previous ones.

1. **Focus on hard examples**
   After each iteration, the weights of misclassified points are increased. This ensures that later weak learners do not simply repeat the same errors but actively try to correct them.
2. **Error-correcting sequence of learners**
   Each weak learner specializes on different problematic regions of the feature space. Collectively, they form a model with significantly reduced bias compared to any single weak learner.
3. **Weighted voting strengthens final predictions**
   Each learner's vote is scaled by its accuracy (e.g., via α in AdaBoost). More accurate learners influence the final decision more strongly, while weaker ones contribute less. This creates a robust aggregate model that outperforms any individual learner.

Thus, boosting's iterative weight updates transform many weak learners—each only slightly better than random—into a **highly accurate and focused ensemble.**

Topic covered	Marks
• States that boosting increases weights on misclassified instances	0.5
• Notes that successive learners focus on previously misclassified samples	0.5
• Explains that this creates an error-correcting chain of weak learners	0.5
• Mentions that weak learners specialize on difficult parts of the data	0.5
• Describes weighted voting enhancing strong learners' impact	0.5
• Concludes that the ensemble becomes stronger than any single weak learner	0.5

4 A.	Differentiate the fundamental assumptions underlying a parametric model (Logistic Regression) from a non-parametric model (Decision Tree) regarding the data generating distribution.	4	Analysis	CO. 5
Ans.				

Topic covered	**Marks**
• Identifies that Logistic Regression assumes a fixed functional form (linear log-odds)	0.5
• Mentions Logistic Regression uses a finite set of parameters independent of dataset size	0.5
• States Logistic Regression assumes a specific distributional structure (sigmoid of linear combination)	0.5
• Notes that Logistic Regression imposes a global model assumption	0.5
• States that Decision Trees make no fixed functional-form assumption	0.5
• Explains that tree complexity grows with dataset size (non-parametric nature)	0.5
• Notes that Decision Trees do not assume any specific probability distribution	0.5
• States that Decision Trees learn local, region-wise boundaries rather than a global structure	0.5

4 B.	Apply the SVM margin (hard margin) formula to compute the margin of a classifier defined by the hyperplane $$3x_1 + 4x_2 - 12 = 0.$$ If the support vectors lie on the parallel lines $$3x_1 + 4x_2 - 6 = 0 \text{ and } 3x_1 + 4x_2 - 18 = 0,$$ Find the margin of the classifier.	3	Apply	CO. 3
Ans.				

We are given:

- Classifier (decision boundary):

$$3\mathrm{x}_1 + 4\mathrm{x}_2 - 12 = 0$$

- Support vector hyperplanes:

$$3\mathrm{x}_1 + 4\mathrm{x}_2 - 6 = 0\text{and}3\mathrm{x}_1 + 4\mathrm{x}_2 - 18 = 0$$

Step 1: Compute ||w||

For a hyperplane

$$w^T x + b = 0, w = (3,4)$$

$$\| w \| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Step 2: Distance between central hyperplane and support hyperplanes

Distance between two parallel hyperplanes

$$\text{distance} = \frac{| b_2 - b_1 |}{\| w \|}$$

Distance from center (–12) to first support line (–6):

$$\frac{| (-12) - (-6) |}{5} = \frac{6}{5} = 1.2$$

Distance from center (–12) to second support line (–18):

$$\frac{| (-12) - (-18) |}{5} = \frac{6}{5} = 1.2$$

Both sides give the same distance, as expected.

Margin of the classifier = 1.2

If needed, the *total margin width* between the support hyperplanes is:

	$$\frac{\mid(-6)-(-18)\mid}{5}=\frac{12}{5}=2.4$$ But the **SVM margin (per side)** is **1.2**. or **2. Calculate the Norm ($\|\mathbf{w}\|$)** The Euclidean norm (magnitude) of the weight vector $\mathbf{w}$ is: $$\|\mathbf{w}\|=\sqrt{w_1^2+w_2^2}$$ $$\|\mathbf{w}\|=\sqrt{(3)^2+(4)^2}$$ $$\|\mathbf{w}\|=\sqrt{9+16}$$ $$\|\mathbf{w}\|=\sqrt{25}$$ $$\|\mathbf{w}\|=5$$ **3. Compute the Margin (γ)** Apply the margin formula $\gamma=\frac{2}{\|\mathbf{w}\|}$: $$\gamma=\frac{2}{5}$$

Topic covered	Marks
Compute magnitude of weight vector	1
Correct use of distance formula between parallel hyperplanes	0.5
Computes distance from hyperplane to support vector line	1
States final margin	0.5

4 C.	Examine the mechanism through which Error-Correcting Output Codes (ECOC) decompose a multiclass classification task into multiple binary subtasks, and analyze the role of the code matrix design in enhancing diversity among the binary learners.	3	Analysis	CO. 6
Ans.	.			

Topic covered	Marks
• States that ECOC uses a code matrix to convert multiclass into multiple binary tasks	0.5
• Explains that each column creates a binary learner by grouping classes into +1 and -1	0.5
• Mentions that decoding compares predicted binary vector with class codewords	0.5
• States that code matrix design controls diversity among learners	0.5
• Mentions that high Hamming distance increases independence/error correction	0.5
• Explains that diverse partitions reduce correlated errors and improve discrimination	0.5

5 A.	You are given the following training data:	5	Apply	CO. 2

RoofType	Location	HouseType
Flat	Urban	Apartment
Sloped	Rural	Cottage
Flat	Suburban	Apartment
Sloped	Rural	Cottage
Sloped	Urban	Villa
Flat	Urban	Apartment

Predict the HouseType for a new property with RoofType = Sloped and Location = Urban. Apply the Naïve Bayes classifier to find the most probable HouseType.

Ans.

Class priors $P(\text{HouseType})$

- Apartment: 3 examples → $P(\text{Apartment}) = 3/6 = 0.5$
- Cottage: 2 examples → $P(\text{Cottage}) = 2/6 = 1/3$
- Villa: 1 example → $P(\text{Villa}) = 1/6$

Likelihoods for each class

For Apartment (3 rows: (Flat,Urban), (Flat,Suburban), (Flat,Urban))

- RoofType:
  - Flat: 3, Sloped: 0→ $P(\text{Sloped} \mid \text{Apartment}) = 0$
- Location:
  - Urban: 2, Suburban: 1, Rural: 0→ $P(\text{Urban} \mid \text{Apartment}) = 2/3$

For Cottage (2 rows: (Sloped,Rural), (Sloped,Rural))

- RoofType:
  - Sloped: 2, Flat: 0→ $P(\text{Sloped} \mid \text{Cottage}) = 1$
- Location:
  - Rural: 2, Urban: 0, Suburban: 0→ $P(\text{Urban} \mid \text{Cottage}) = 0$

For Villa (1 row: (Sloped,Urban))

- RoofType:
  - Sloped: 1→ $P(\text{Sloped} \mid \text{Villa}) = 1$
- Location:
  - Urban: 1→ $P(\text{Urban} \mid \text{Villa}) = 1$

Posterior (up to proportionality) for new property

New: RoofType = **Sloped**, Location = **Urban**

We compute:

$$P(C \mid \text{Sloped,Urban}) \propto P(C)\, P(\text{Sloped} \mid C)\, P(\text{Urban} \mid C)$$

Apartment

$$P(\text{Apartment}) \cdot P(\text{Sloped} \mid \text{Apartment}) \cdot P(\text{Urban} \mid \text{Apartment}) = 0.5 \cdot 0 \cdot \frac{2}{3} = 0$$

Cottage

$$P(\text{Cottage}) \cdot P(\text{Sloped} \mid \text{Cottage}) \cdot P(\text{Urban} \mid \text{Cottage}) = \frac{1}{3} \cdot 1 \cdot 0 = 0$$

Villa

$$P(\text{Villa}) \cdot P(\text{Sloped} \mid \text{Villa}) \cdot P(\text{Urban} \mid \text{Villa}) = \frac{1}{6} \cdot 1 \cdot 1 = \frac{1}{6}$$

Only **Villa** has a non-zero posterior.

Final prediction:

$$\boxed{\text{HouseType} = \text{Villa}}$$

Topic covered	Marks
Identifying class frequencies correctly • Correct priors	0.5
Correct likelihoods for Apartment • P(Sloped ∣ Apartment) • P(Urban ∣ Apartment)	0.5
Correct likelihoods for Cottage • P(Sloped ∣ Cottage) • P(Urban ∣ Cottage)	0.5
Correct likelihoods for Villa • P(Sloped ∣ Villa) • P(Urban ∣ Villa)	0.5
Posterior probability • Correct formula replacement for atleast 2 • Correct posterior probability	0.5*2=1 0.5*3=1.5
Clearly states final predicted HouseType = Villa	0.5

5 B.	Determine the principle of the ε-insensitive loss function in Support Vector Regression (SVR), and examine the role of the parameter ε in defining the regression boundary, determining the set of support vectors, and influencing model sparsity. Further, analyze the trade-offs introduced by different ε values in balancing precise data fitting with robustness and tolerance to noise.	5	Analysis	CO. 3

Ans.

Topic covered	Marks
Principle of ε-insensitive Loss • Correct description of ε-insensitive loss (zero loss within ε tube) • Explanation that only deviations beyond ε incur penalty	0.5 0.5
Role of ε in Regression Boundary • **ε defines width of ε-tube** • **Smaller ε → tighter fit; larger ε → looser fit**	0.5 0.5
Role of ε in Support Vector Determination • Points outside tube become support vectors • ε controls # of SVs (larger ε → fewer; smaller ε → more)	0.5 0.5
Influence on Sparsity • Fewer SVs produce sparse model • Larger ε increases sparsity; smaller ε reduces sparsity	0.5 0.5
Trade-Offs of Different ε Values • Small ε → precise but noisy/overfitting • Large ε → robust but may underfit	0.5 0.5